

maxis 4G

- Full Sensitivity Resolution - Extreme Performance

Innovation with Integrity

UHR-TOF MS

Premium Power and Performance



maXis™ UHR-TOF technology has set the standard in high-resolution tandem mass spectrometry across a wide range of analytical applications.

The maXis 4G, with Full Sensitivity Resolution, continues to advance the performance level and capabilities of this revolutionary series of instruments. With significant enhancements in mass accuracy and resolution, and unparalleled sensitivity and speed, the maXis 4G offers the confidence to measure and identify small molecules, proteins and intact large molecules, such as antibodies, with unprecedented levels of certainty in both MS and MS/MS modes.

Redefining high performance mass spectrometry

- Outstanding Mass Resolution and Accuracy in all Modes
- Full Sensitivity Resolution Technology
- Complete Flexibility to Analyze Large and Small Molecules
- Compatible with Fast U-HPLC Separations
- Intuitive Instrument Control and Data Analysis Software
- Robust, Stable, and Easy to Maintain Platform

"The four maXis instruments in our laboratory are essential components of our total mass spectrometry infrastructure. For metabolomics studies, dedicated GC-, CE-, and LC-platforms benefit from the high resolution and mass accuracy provided by the maXis. In addition, one instrument plays a pivotal role in proteomics applications. The exchange of all our maXis instruments by the newly developed maXis 4G is expected to further improve the data quality needed for our clinical metabolomics and proteomics studies."

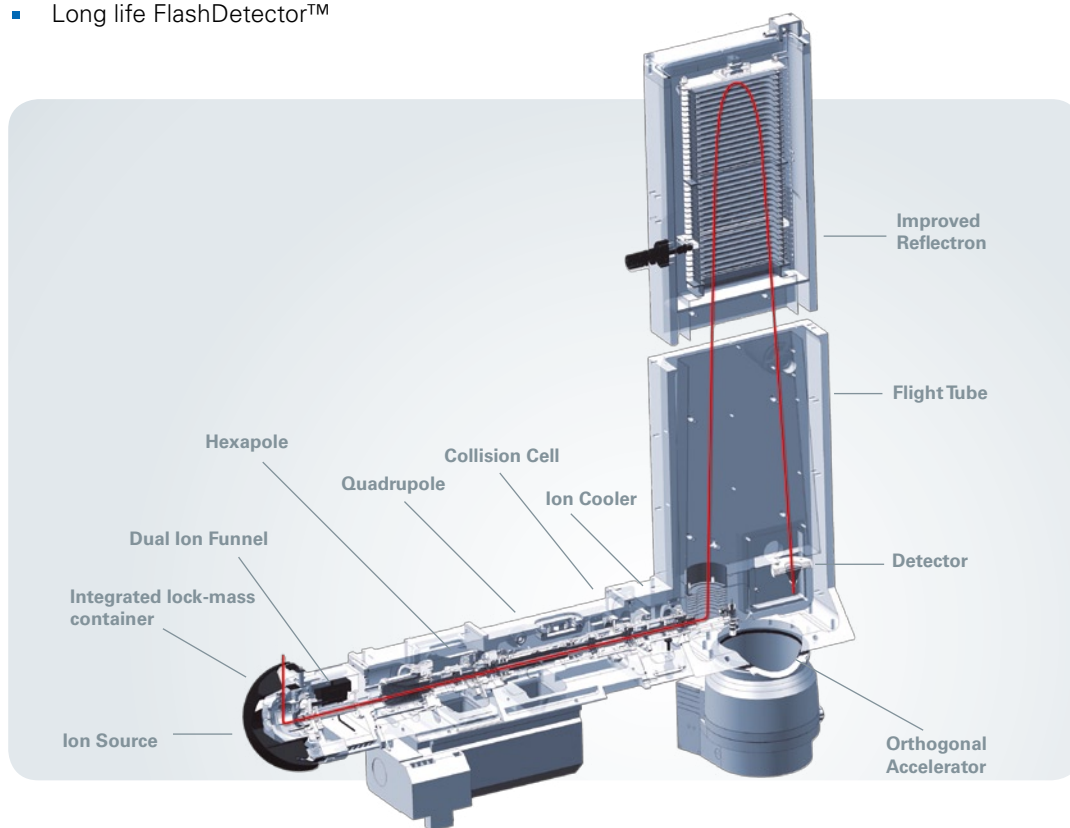
Professor Andre M. Deelder, Head of Parasitology & Biomolecular Mass Spectrometry Units, Leiden University Medical Centre, The Netherlands



● Advancing Mass Spectrometry Beyond the Expected

No one element enables the performance of the maXis UHR-TOF series. A number of fundamental discoveries and innovations in Mass Spectrometry instrument design and engineering have led to the development of this groundbreaking technology.

- 3rd Generation Ion Cooler™
- Integrated lock-mass device delivering for reliable and exceptional internal calibration
- Enhanced performance Reflectron design delivering exceptional Full Sensitivity Resolution
- 4 GHz Sample digitizer
- Long life FlashDetector™



maXis 4G - the Essential System for:

- Small Molecule Identification
- Protein Characterization
- Quantitative Proteomics
- Biomarker Discovery

Readily Equipped for Key Analytical Tasks

- Molecular Formula Determination
- Top-down Characterization of Intact Proteins
- Top-down Profiling and Quantitation
- Bottom-up Studies of Peptides and PTMs
- PTM-site Determinations

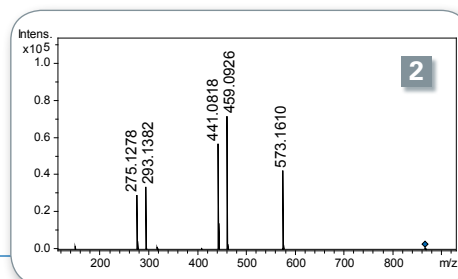
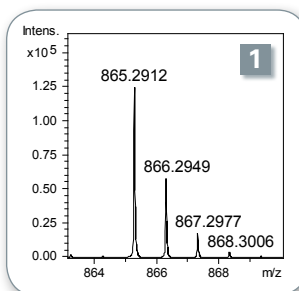
Power for a Breadth of Applications

- Intact Protein Analysis
- Synthetic or Natural Compound Molecular Formula Determination
- Drug and Metabolite Identification
- Impurity and Degradant Identification
- Biopharmaceutical QC
- Metabolomics Studies
- Pesticide or Contaminant Testing
- Forensic Drug and Metabolite Testing
- Lipid and Glycolipid Analysis
- Carbohydrate and Glycan Analysis
- Protein and Peptide Identification and Characterization

Certainty in Small Molecule Analysis

Fast and Accurate Identification of Unknowns

For molecular identifications of unknowns, maXis 4G MS/MS data quality, in combination with Bruker's unique *ab initio* formula discovery tool SmartFormula 3D™, rapidly and precisely identifies molecular species. This integrated combination of hardware and software uses accurate mass and isotopic analysis of MS and MS/MS results, without assumptions or reference to libraries, to provide very accurate molecular formulae with a maximum level of certainty.



Sum formula generation with SmartFormula3D

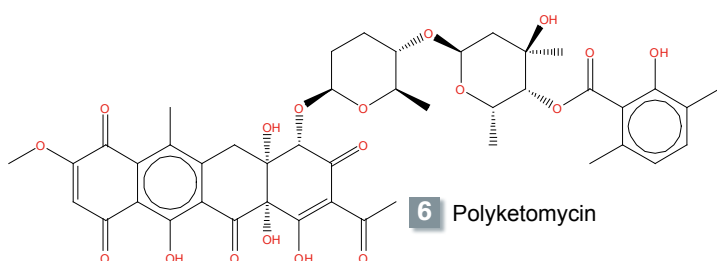
SumFormula	m/z calc	err[mDa]	err[ppm]	mSigma	eConf	Comb.mSigma
☒ C ₄₄ H ₄₉ O ₁₈	865.2913	0.1	0.2	10.1	even	11.6

SumFormula	SumFormula Loss	m/z Loss	Delta SF	m/z calc
☐ C ₂₈ H ₂₉ O ₁₃	C ₁₆ H ₂₀ O ₅	292.1302	C ₁₂ H ₁₀ O ₉	573.1603
☐ C ₂₂ H ₁₉ O ₁₁	C ₂₂ H ₃₀ O ₇	406.1986	C ₆ O ₇	459.0922
☐ C ₂₂ H ₁₇ O ₁₀	C ₂₂ H ₃₂ O ₈	424.2094		441.0816
☐ C ₁₆ H ₂₁ O ₅	C ₂₈ H ₂₈ O ₁₃	572.1530	H ₂ O	293.1384
☐ C ₁₆ H ₁₉ O ₄	C ₂₈ H ₃₀ O ₁₄	590.1634		275.1278

#	Compound	Compound ID	Database
1	1	found nothing	HMDB
2	2	found nothing	METLIN
3	[(2S,3R,4R,6S)-6-[(2R,3S,6S)-6-[(1S,...	4677656	ChemSpider

When asked about the impact the maXis and Smartformula 3D has had in Structure Elucidation at Pfizer Global Research & Development, Dr Don Richards said:

"This is the advance in mass spec technology that we've all been waiting for! It's the first time that we've really been able to use 5 decimal places for mass accuracy and it has already greatly enhanced our productivity."



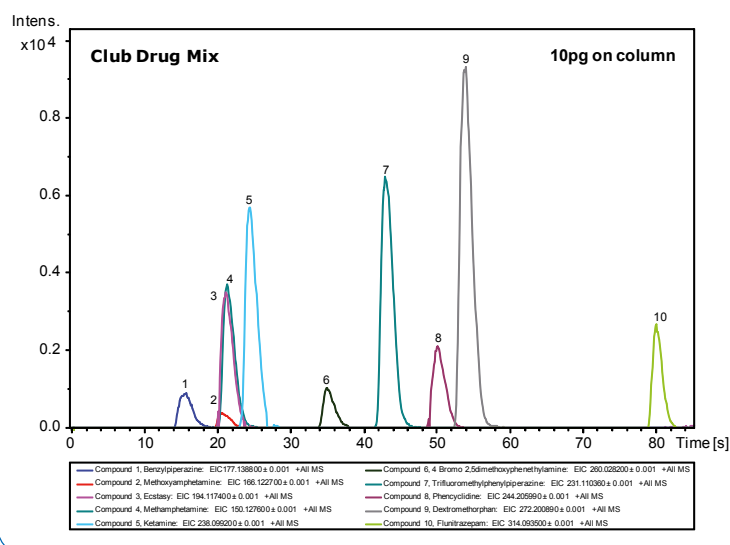
6 Polyketomycin

Structure elucidation of Polyketomycin, a natural product from microorganisms. (1) The isotopic pattern of Polyketomycin, all isotopes showing a mass accuracy of better than 1 ppm. (2) Fragmented spectrum. The information of MS and MS/MS spectra were used for sum formula generation with SmartFormula3D™. Only one possible elemental composition remains. (3) SmartFormula also explains the sum formulae for all fragment ions. (4) A database search revealed Polyketomycin as the correct hit, here using IUPAC naming (5 and 6). Sample kindly provided by BioVotica GmbH, Göttingen, Germany.

Keeping Pace with Modern Separations

Increased chromatography speeds need to be matched by appropriate mass spectrometry capabilities. The maXis 4G is completely compatible with even the fastest separations. Capable of acquisition speeds of up to 30 Hz, the maXis 4G provides the rate of measurement necessary to obtain excellent quality data from accelerated analyses and the resulting very narrow sample peaks.

High speed compound ID of mixtures



High resolution extracted ion chromatograms of pharmaceutical compounds at an acquisition speed of 10 spectra per second analyzed in under 85 seconds.

Metabolomics

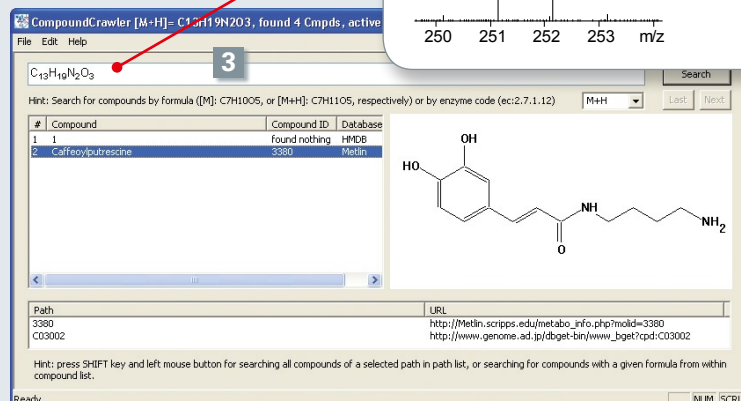
Apart from statistical analysis, the fast and easy identification of relevant features is one of the key requirements of metabolomics studies. The unique UHR-TOF technology makes the maXis 4G an ideal metabolic profiling platform with outstanding resolution and high mass accuracy for compound identification.

Data kindly provided by Prof. I. T. Baldwin, Dr. M. Schöttner, Max Planck Institute for Chemical Ecology, Jena, Germany

Further Reading

Development and validation of a liquid chromatography-electrospray ionization-time-of-flight mass spectrometry method for induced changes in *Nicotiana attenuata* leaves during simulated herbivory. E. Gaquerel, S. Heiling, M. Schoettner, G. Zurek, I.T. Baldwin
J Agric Food Chem. 2010 Sep 8;58(17):9418-27.

Time dependent metabolic changes in *Nicotiana attenuata* after simulated insect herbivory



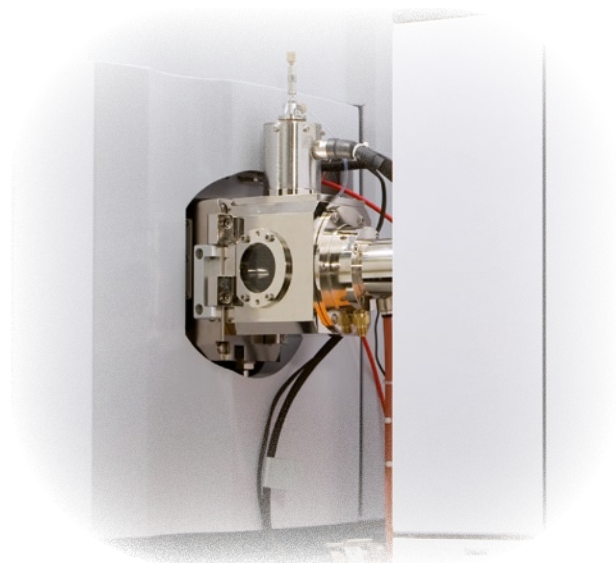
1. Scores and Loadings Plot. 2. maXis 4G MS spectrum and SmartFormula result of highlighted loading. 3. Compound Crawler database query for fast & easy web-based identification e.g. in ChemSpider database. Search result for [M+H]⁺ = C₁₃H₁₉N₂O₃ is Caffeoylputrescine.

Expanded Capabilities With GC and Direct Probe Options

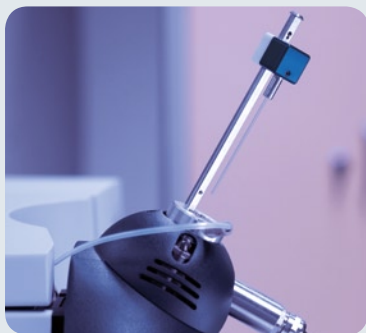
GC-APCI

A new GC-APCI source enables the combination of UHR-TOF MS with both LC and GC.

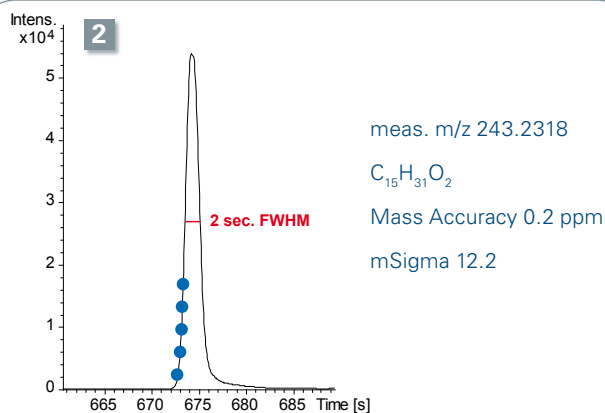
The maXis 4G UHR-TOF system provides exact mass accuracy and resolving power to expand the horizon for GC-MS based analyses. Data acquisition rates of 20Hz and faster are mandatory for typical GC peak widths of < 2 s (FWHM). The analysis of Fatty Acid methyl esters (FAME), shown below, shows that the maXis 4G has the capabilities to deliver high mass accuracy and resolution even at extreme scan speeds. The unambiguous formula ID from the GC-TOF-MS run is determined using SmartFormula™.



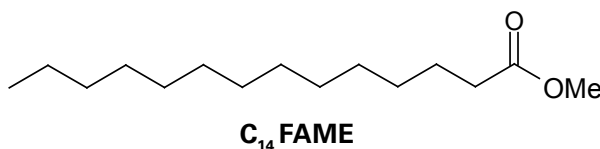
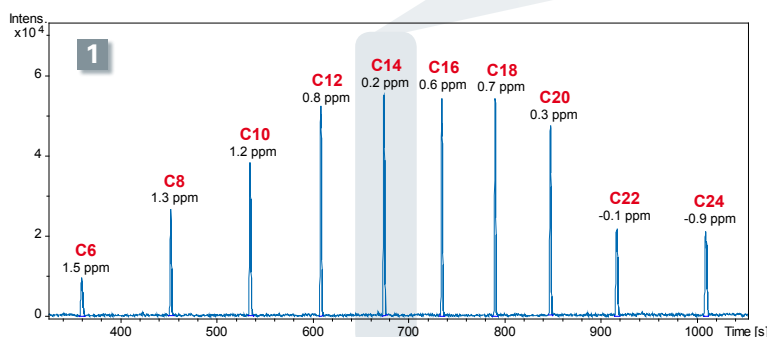
Direct Probe For Solid Samples



The direct probe ion source allows straight-forward analysis of liquid, solid or insoluble samples.



Mass Accuracy at GC-speed allows for Formula ID



GC-TOF-MS analysis of FAME. 1. Base Peak Chromatogram of 0.5 ng C6 – C24 FAME. Indicated are mass accuracy in ppm and the fatty acid chain length. 2. EIC Detail of C14 FAME peak clearly demonstrating the capability of maXis to deliver 1 mDa traces even at 20 Hz scan speed.

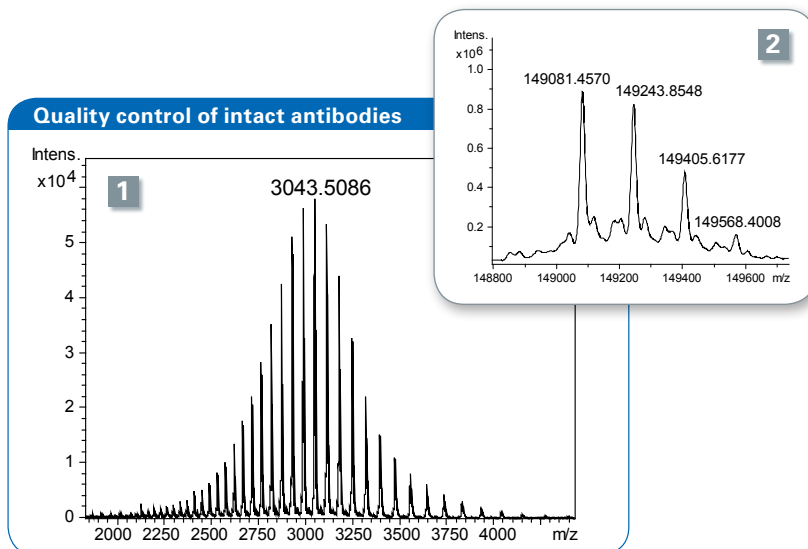
● Protein Identification and Characterization

Biopharma QC: Intact Protein Analysis

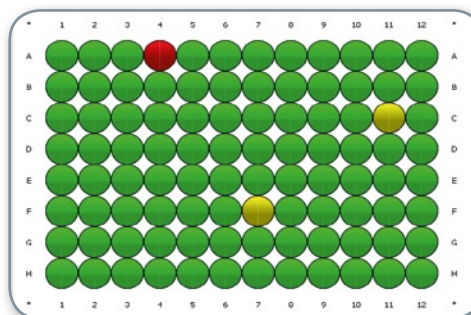
Due to its superior resolution, in the high mass range, the maXis 4G is the ideal platform for the analysis of intact proteins, regardless of their size. Small and medium size proteins can be measured isotopically resolved. Larger proteins, like antibodies, give excellent charge distribution patterns which are deconvoluted by using the Maximum Entropy algorithm. In combination with data from protein digests, complete characterization including PTMs can be achieved.

Protein Identification and Quantitation: Labeled & Label-free

The maXis 4G provides increased intensity, stability, mass accuracy, and resolving power for protein analysis. These features can lead to very accurate and precise quantitation results for proteins found in complex biological mixtures. Utilization of the Proteinscape™ software platform for data-dependent and targeted MS/MS precursor selection can also increase the number of identified, quantified, and validated peptides and proteins in samples.

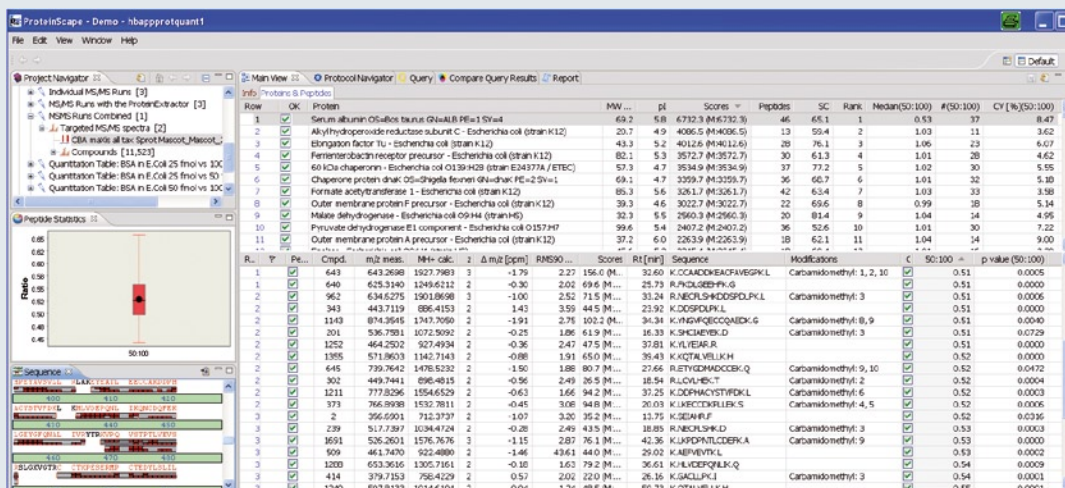


An IgG96 was desalted by using a C8 HPLC column giving a mass spectrum with an charge envelope between m/z 2000 and 4500 (1). Charge deconvolution with maximum entropy gives the masses of the intact antibody and several glycoforms (2).



Overview of BioPharma Compass Software: the software can be completely automated and set up to compare samples with reference standards. Individual QC criteria are user defined and reports are automatically generated.

ProteinScope supports labeling and label-free quantitation techniques



Technical Specifications

Cutting-Edge Performance

- First ever ultra-high resolution time-of-flight technology; UHR-TOF
- Outstanding combination of mass accuracy, resolution and speed without compromise
- Dual ion funnel Q-q front end
- SmartFormula 3D, the novel combination of accurate mass of parent and fragment ions with True Isotopic Pattern (TIP™)
- Wide dynamic range for ultra-stable accurate mass
- IonCooler™ technology for full MS/MS bandwidth
- Dimensions: 800 x 1320 x 2850 mm, weight 500 kg

Source Options

- APCI (atmospheric pressure chemical ionization) source
- APPI (atmospheric pressure photo ionization) source
- Direct probe option
- Online / offline NanoElectrospray sources
- CE/MS coupling with grounded ESI needle
- Direct GC coupling
- CryoSpray Source

Analytical Performance

- Mass accuracy: 600ppb RMS Error
- High resolution EIC with typically a 1 mDa window
- Mass resolution: 60,000 FSR at UHPLC speed
- Mass range 50 – 20,000 m/z
- Advanced temperature compensation
- Acquisition rate up to 30 Hz (profile and peak detected spectra to disk)

Compass & Application Software Suites

- Integrated LC-MS/MS control and data processing incl. SmartFormula 3D
- MetaboliteTools™ for metabolite identification
- TargetAnalysis™ for compound screening
- ProteinScape™ - the bioinformatics platform
- Compass OpenAccess™: Walk-up LC-MS chemical formula generation
- ProfileAnalysis™ for LC-MS based profiling and label-free quantitation

Support of HPLC and sample inlet systems from the following vendors:

Bruker EASY-nLC, Advion TriVersa NanoMate, Agilent, Dionex, VWR/Hitachi, Waters (incl. UPLC), Autosamplers from CTC, Michrom

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